

Assignment for Transport Phenomena II (Self-Study Week 2)

Assignment Title: Mass Transfer in Everyday Life: Volatile Liquid Evaporation and Analysis

Objective: Observe and analyze mass transfer phenomena in real life by studying the evaporation of a volatile liquid and applying transport principles to calculate key parameters.

Task:

1. Experiment Setup:

- Select a volatile liquid (e.g., ethanol-based hand sanitizer, alcohol solution, or similar household liquid).
- Pour a known amount into a container and expose it to ambient conditions.
- Record:
 - Initial liquid volume or mass
 - Ambient temperature and airflow conditions
 - Time intervals for noticeable changes (e.g., every 5 minutes until significant evaporation occurs)
- Take clear photos of the setup and measurement process.

2. Observations:

- Track the mass or volume loss over time.
- Note environmental conditions (temperature, humidity, airflow).
- If possible, repeat under two conditions:
 - Natural evaporation (still air)
 - Forced convection (e.g., by using a fan)

3. Analysis:

- Calculate:
 - a) Diffusion coefficient and diffusion flux of the volatile component in air (based on your data and assumptions).
 - b) Convective mass transfer coefficient under natural and forced convection.
- Estimate:
 - Time required for complete evaporation and evaporation rate under both conditions.
- Clearly state all assumptions, including:
 - Physical properties (density, vapor pressure, diffusivity)
 - Ambient conditions
 - Geometry simplifications
 - Neglecting minor effects (e.g., radiation, turbulence beyond fan effect)

(You are expected to choose appropriate methods and formulas based on what you learned in class.)

4. Report Format:

- Prepare a PowerPoint presentation including:
 - Photos of the experiment
 - Description of the setup and procedure
 - Assumptions and reasoning
 - Data tables and graphs (time vs. mass/volume)
 - Calculations and analysis
 - Discussion of results and observations
 - Conclusion: What did you learn about mass transfer in real life?
- Save the PPT as PDF and submit online.

Evaluation Criteria:

- Creativity & clarity of experiment design
- Correct application of mass transfer principles
- Depth of analysis & discussion
- Presentation quality (organization, visuals, clarity)